VyapaarSetu

व्यापार सेतु — Trade Bridge

AI-Powered MSE Agent Mapping Tool for ONDC TEAM Initiative

IndiaAI Innovation Challenge 2026

Problem Statement 2 — Ministry of MSME

Ministry of Micro, Small and Medium Enterprises • NSIC • MeitY

TEAM Scheme: Ambitious Goal, Manual Execution

5L	277 Cr	50%	14 Days
MSEs to Onboard onto ONDC	TEAM Scheme Budget (3 Years)	Women-Led MSE Target	Current Avg. Onboarding Time

1. **Manual data entry barrier** — MSEs fill complex digital forms they don't understand; rural artisans can't navigate English ONDC portals
2. **Inconsistent product categorisation** — Wrong ONDC taxonomy = products invisible on buyer apps; no standardised CaaS layer exists
3. **Labour-intensive NSIC verification** — Manual claim review creates delays, rejections, bottlenecks at scale
4. **No intelligent MSE-SNP matching** — Random SNP allocation ignores category fit, geography, and capacity — leading to poor outcomes

Result: At current pace, TEAM scheme will miss its 5 lakh target. Manual processes cannot scale to India's 6.3 Cr MSMEs.

OUR SOLUTION

VyapaarSetu — AI Agent That Onboards MSEs onto ONDC

VyapaarSetu is an AI-powered platform that automates the entire MSE onboarding journey — from voice-based registration to ONDC-compliant catalogue building to intelligent SNP matching — eliminating manual data entry and language barriers.

1. Voice-Enabled Registration

MSE speaks in Hindi/Hinglish on phone or web. AI extracts business details, auto-fetches Udyam data, fills all registration fields. No forms, no typing.

2. AI Product Cataloguing (CaaS)

MSE describes products in natural language or sends WhatsApp photo. AI classifies to correct ONDC RET taxonomy, extracts mandatory attributes, validates compliance.

3. Intelligent SNP Matching

AI scores and ranks SNPs by category overlap, geography, B2B/B2C fit, and capacity. MSE selects best-fit SNP. Onboarding package auto-generated.

MSE Calls Voice / WhatsApp → **AI Registers** Udyam Auto-Fill → **AI Catalogues** ONDC Taxonomy → **AI Matches** Best-Fit SNP → **Live on ONDC** All Buyer Apps

Key Differentiator: Zero digital literacy required. An MSE owner can onboard onto ONDC by simply making a phone call — no internet, no smartphone, no forms.

Three AI Capabilities, One Seamless Journey

Voice-First Registration

- Web Speech API + Sarvam AI STT/TTS
- Hindi, English, Hinglish supported
- Udyam number → auto-fetch business details
- LLM extracts structured data from speech
- Production: toll-free IVR (no internet needed)

Intelligent SNP Matching

- Weighted multi-factor scoring algorithm
- Category overlap (45%) + Geography (25%)
- Transaction type B2B/B2C (20%) + Capacity (10%)
- Qdrant vector similarity on SNP profiles
- Top 3 recommendations with match reasons

AI Cataloguing-as-a-Service (CaaS)

- Natural language → ONDC RET code classification
- NER extracts: name, MRP, MOQ, HSN, material, origin
- E-Commerce Rules 2020 compliance validation
- 10 ONDC categories (RET10-RET19) supported
- WhatsApp follow-up for missing fields

Multi-Channel AI Agent

- Skill-based ReAct agent with 8 specialised tools
- 3 skills: Onboarding, Cataloguing, Matching
- Web chat + Voice + WhatsApp channels
- Voice → WhatsApp handoff for catalogue completion
- RAG over TEAM scheme + ONDC knowledge base

TECHNOLOGY

Architecture & Tech Stack

Layer	Technology
Frontend	Next.js 16 + React 19 (3 portals)
Backend	FastAPI (async Python)
AI/LLM	DeepSeek via LangChain
Voice	Sarvam AI STT/TTS + Web Speech API
Embeddings	sentence-transformers (local)
Database	PostgreSQL + SQLAlchemy async
Vector DB	Qdrant (RAG knowledge base)
Cache	Redis (sessions, state)
Messaging	WhatsApp via Baileys
Auth	JWT (python-jose)
Deploy	Docker Compose

System Architecture

```
MSE (Voice/Chat/WhatsApp)
       |
       v
Next.js Frontend (3 Portals)
       |
       v
FastAPI Backend + AI Agent Engine
  |         |         |
  v         v         v
PostgreSQL   Redis     Qdrant
(MSE, SNP,  (sessions) (ONDC taxonomy
 products)             + TEAM RAG)
       |
       v
DeepSeek LLM (LangChain Agent)
```

3 Portals: MSE (onboarding), SNP (lead management), NSIC Admin (verification + analytics)

PLATFORM

Three Portals, One Ecosystem

MSE Portal

PRIMARY — Competition Focus

- **Landing page** — TEAM scheme explainer
- **Voice registration** — Udyam + profile
- **AI catalogue builder** — web + voice + WhatsApp
- **SNP matching** — top 3 recommendations
- **Dashboard** — onboarding status tracker
- **Knowledge base** — TEAM/ONDC FAQs (RAG)

SNP Portal

Seller Network Participants

- **Dashboard** — stats overview
- **AI-matched leads** — MSEs with readiness score
- **Lead detail** — catalogue preview + compliance
- **Accept/Reject** — one-click MSE onboarding
- API integration for bulk catalogue import

NSIC Admin Portal

Verification & Analytics

- **Funnel analytics** — registered → matched → live
- **AI verification queue** — pre-screened MSEs
- **WhatsApp bot management** — connect/monitor
- **State-wise reporting** — TEAM scheme KPIs
- Reduces manual verification workload by 80%

MSE Registers Voice / WhatsApp → **AI Catalogues** CaaS Layer → **AI Matches SNP** Top 3 Ranked → **SNP Accepts** Lead Review → **NSIC Verifies** AI Pre-Screened → **Live on ONDC** All Buyer Apps

Massive Addressable Market, Zero Competition

TAM — Total Addressable Market
6.3 Cr registered MSMEs in India
6.3Cr

SAM — TEAM Scheme Target
5 lakh MSEs to onboard onto ONDC
5L

SOM — Our Target (2 Years)
2 lakh MSEs (40% market share)
2L

Competitive Landscape

Approach	Limitation
Manual TEAM camps	Doesn't scale; 14-day avg onboarding
Unicommerce / Vinculum	Generic e-com; no ONDC taxonomy
SNP self-onboarding	English-only portals; no voice
Generic chatbots	No CaaS layer; no compliance check

Our moat: Only solution combining voice-first registration + ONDC CaaS + intelligent SNP matching in one platform. No competitor addresses all three PS2 requirements.

Government SaaS (GovTech) + SNP Transaction Fees

Government Contract Revenue

- **Stage 1:** INR 25 Lakhs (development + pilot)
- **Stage 2:** INR 1 Crore (2-year deployment contract)
- Funded by TEAM scheme (Rs 277 Cr budget)

Per-MSE Onboarding Fee

- Rs 50-100 per MSE charged to SNPs
- Value: pre-catalogued, compliance-checked leads
- 2L MSEs x Rs 75 avg = Rs 1.5 Cr revenue potential

Premium CaaS API Access

- SNPs pay for bulk catalogue enrichment API
- Priority matching and lead placement
- Annual subscription model

Unit Economics

Metric	Value
Cost per MSE onboarding	~Rs 15 (LLM + infra)
Revenue per MSE (SNP fee)	Rs 50-100
Gross margin per MSE	~75-85%
Year 1 target MSEs	50,000
Year 2 target MSEs	2,00,000

Zero cost to MSEs. Platform is free for MSE sellers throughout — revenue comes from government contract and SNP transaction fees. Aligned with TEAM scheme's MSE-first mandate.

Three-Phase Rollout via NSIC Network

PHASE 1 • MONTHS 1-6

Pilot

- 5 high-MSME states
- Maharashtra, Gujarat, Rajasthan, UP, Tamil Nadu
- 500 MSEs via NSIC district camps
- DIC office integration
- 6 SNP partners onboarded

500
MSEs Target

PHASE 2 • MONTHS 7-12

Scale

- Pan-India via NSIC 600+ branches
- Toll-free IVR for rural MSEs
- WhatsApp Business API upgrade
- Regional language expansion
- CSC network partnership

50K
MSEs Target

PHASE 3 • MONTHS 13-24

National

- Full TEAM portal integration
- MSME ministry SMS/notification push
- 20+ SNP integrations
- All 36 states/UTs covered
- 50% women-led MSE focus

2L
MSEs Target

Distribution advantage: NSIC's 600+ branch network + 4 lakh CSCs + MSME ministry notification channels = built-in acquisition funnel at zero CAC.

Key Performance Indicators

AI Performance Metrics

Metric	Target
ONDC classification accuracy	> 92%
Attribute extraction F1	> 88%
SNP match acceptance rate	> 70%
Catalogue compliance score	> 85%
Voice STT accuracy (Hindi)	> 90%
Agent response latency	< 3 seconds

Impact Metrics

Metric	Before	After
Onboarding time	14 days	< 3 days
Form completion rate	35%	> 85%
NSIC verification time	5 days	< 1 day

Regulatory Compliance

DPDP Act 2023

Consent-based data collection, purpose limitation, data minimisation, user right to erasure

E-Commerce Rules 2020

Mandatory product attribute validation — MRP, country of origin, seller details on every listing

Legal Metrology Act

Weight/volume declarations, net quantity, packaging date compliance checks

Responsible AI Guidelines

Transparent AI decisions, human-in-loop verification, bias monitoring across categories

Cybersecurity Framework

JWT auth, encrypted data at rest/transit, rate limiting, input sanitisation

Partnerships & Collaborations

Government Partners

- **NSIC** — implementing agency, 600+ branch distribution network
- **Ministry of MSME** — policy alignment, Udyam database access
- **District Industries Centres** — ground-level MSE acquisition

Technology Partners

- **ONDC** — network integration, taxonomy standards
- **Sarvam AI** — Indic language STT/TTS
- **DeepSeek** — LLM backbone for AI agents

SNP Partners (6 Live on ONDC)

SNP	Categories	B2B/B2C
GlobalLinker	RET10,12,13,14,15,16,18	Both
Plotch	RET10,12,13,14,15,16,18	B2C
SellerApp	RET10,12,14,16	Both
eSamudaay	RET11	B2C
DotPe	RET10,11,12,13,16	B2C
MyStore	RET10,12,13,14,16	B2C

Potential Partners

- **CSC e-Governance** — 4 lakh+ Common Service Centres
- **Twilio/Exotel** — telephony infrastructure
- **Meta (WhatsApp Business)** — messaging at scale

VyapaarSetu

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Bridging 5 Lakh MSEs to ONDC — One Voice Call at a Time

Voice-First
No literacy barrier

AI CaaS
ONDC taxonomy compliance

Smart Match
Best-fit SNP allocation

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Thank You